

Web Application Development

❖ Bootstrap CSS

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Outlines

- Bootstrap CSS
- Page layouts using bootstrap CSS

What is bootstrap CSS?

- **Bootstrap** is a free and open-source front-end framework for designing websites and web applications.
- **Bootstrap** is a free front-end framework for faster and easier web development that helps you to create a responsive front-end design.

Why use Bootstrap?

- Saves time (no need to write CSS from scratch)
- Built-in responsiveness
- Consistent design

Why we need to use Bootstrap?

- **Easy to use:**
 - Anybody with just basic knowledge of HTML and CSS can start using Bootstrap
- **Responsive features:**
 - Bootstrap's responsive CSS adjusts to phones, tablets, and desktops
- **Mobile-first approach:**
 - In Bootstrap 3, mobile-first styles are part of the core framework
- **Browser compatibility:**
 - Bootstrap is compatible with all modern browsers (Chrome, Firefox, Internet Explorer, Edge, Safari, and Opera)

How to design a page in bootstrap CSS?

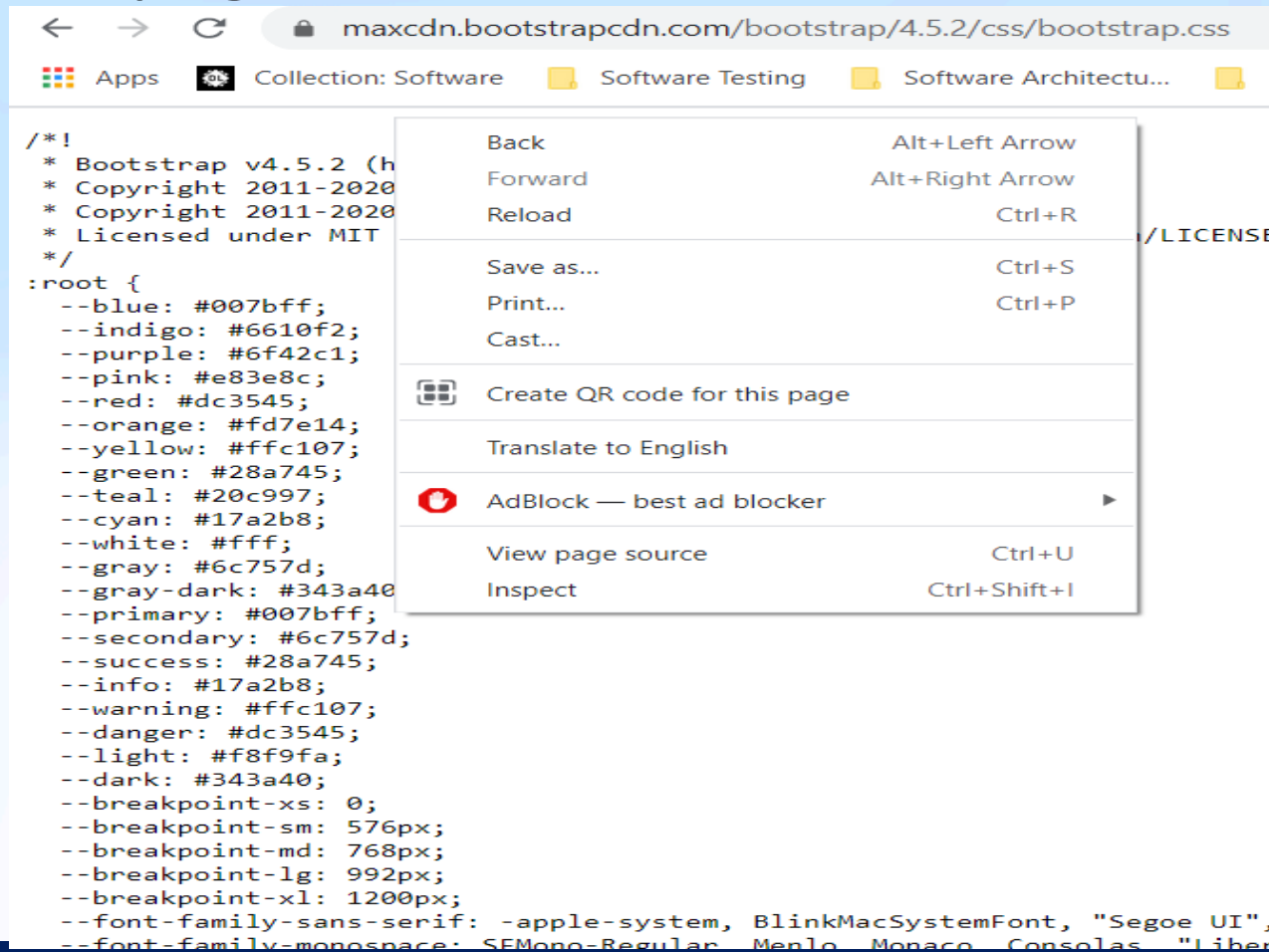
- Download bootstrap css file
 - There are many sources
 - <https://bootswatch.com/>
 - <https://maxcdn.bootstrapcdn.com/bootstrap/4.5.2/css/bootstrap.min.css>
 - <https://getbootstrap.com/>

Learning Resources

- <https://www.tutorialrepublic.com/twitter-bootstrap-tutorial/>

Walkthrough

- Download bootstrap css form web source and add it in your html page



Walkthrough

- Create a simple web page that should display a table

```
<head>
  <title>Bootstrap Example</title>
  <link rel="stylesheet" href="bootstrap.css">
</head>
<body>
  <h2>Basic Table</h2>
  <!-- to decorate a table, you have to use .table class -->
  <table class="table">
```

Basic Table

Muhammad	Asad	asad@example.com
Numan	Saleem	numan@example.com
Khalid	Naeem	khalid@example.com

```
</tr>
<tr>
  <td>Khalid</td>
  <td>Naeem</td>
  <td>khalid@example.com</td>
</tr>
</table>
</body>
</html>
```

Bootstrap selectors

- Bootstrap contains wide range of selectors. Some very common selectors in bootstrap are as following
 - Bootstrap Grid System
 - Bootstrap typography
 - Bootstrap Tables
 - Bootstrap buttons
 - Bootstrap Pagination
 - Bootstrap Navbar

Grid layouts

- In order to make a grid layout, a developer need to have knowledge about following
 - Rows
 - Row is a class that creates new row.
 - Everything (tags) inside row accommodate themselves in horizontal order
 - Columns
 - Column is a class that creates column
 - Column are vertically ordered
 - If columns are child of row, then the order becomes horizontal

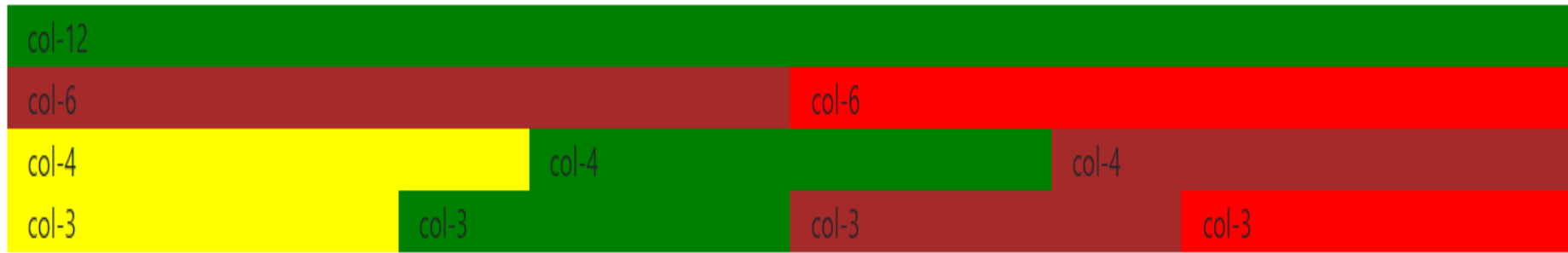
Grid layout

- There are multiple classes of column. For example
 - Col-xl-12, col-xl-11, col-xl-10,...col-xl-1
 - Creates columns on extra large display (desktops)
 - Col-lg-12, col-lg-11, col-lg-10,...col-lg-1
 - Creates columns on large display (laptops)
 - Col-md-12, col-md-11, col-md-10,...col-md-1
 - Creates columns on medium display (tablets)
 - Col-sm-12, col-sm-11, col-sm-10,...col-sm-1
 - Creates columns on small display (mobile)

Column Example

The following code summarizes how the Bootstrap grid system works across multiple devices: (See Next page)

column Demo



```
</div>
<div class="col-lg-4" style="background-color:yellow">
  col-4
</div>
<div class="col-lg-4" style="background-color:green">
  col-4
</div>
<div class="col-lg-4" style="background-color:brown">
  col-4
</div>
<div class="col-lg-3" style="background-color:yellow">
  col-3
</div>
<div class="col-lg-3" style="background-color:green">
  col-3
</div>
<div class="col-lg-3" style="background-color:brown">
  col-3
</div>
<div class="col-lg-3" style="background-color:red">
  col-3
</div>
```

Column example

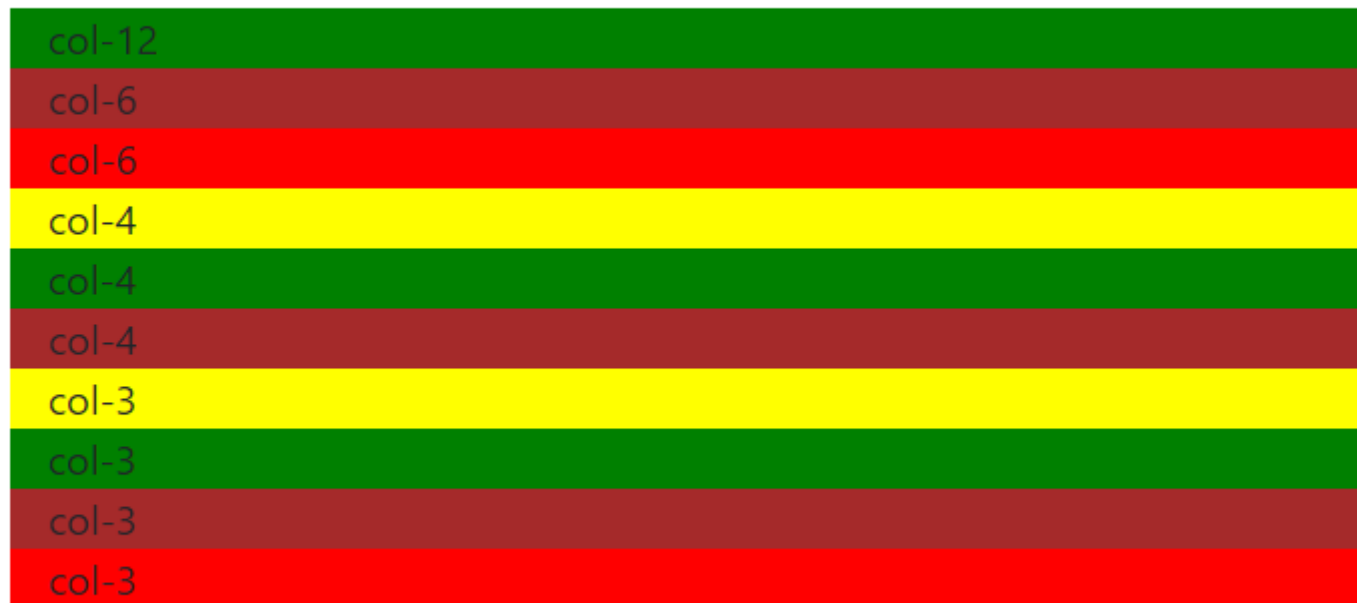
- However, if you minimize the browser size, the column property will get exploit, and the previous code will shows each column in a form of row

```

<body>
  <div class="container">
    <h2>column Demo</h2>
    <div class="row">
      <div class="col-lg-12" style="background-color:green">
        col-12

```

column Demo



```

    </div>
    <div class="col-lg-3" style="background-color:brown">
      col-3
    </div>
    <div class="col-lg-3" style="background-color:red">
      col-3
    </div>
  </div>
</body>

```

Column Example

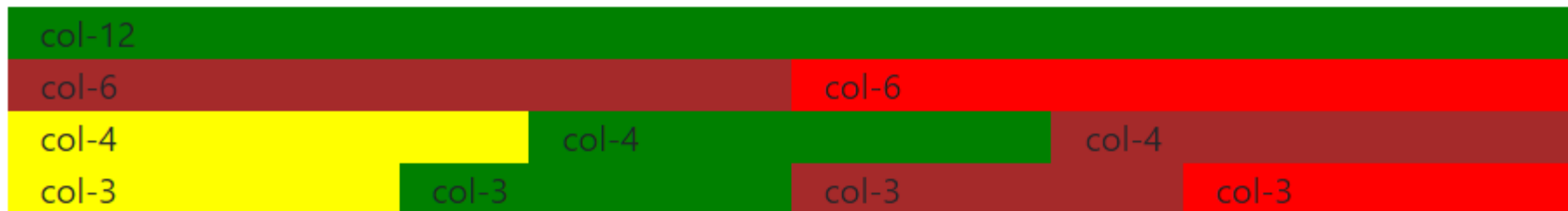
- However, if you still want to maintain the original display on medium size browser, then use col-md property. For example (next slide)

```

<body>
  <div class="container">
    <h2>column Demo</h2>
    <div class="row">
      <div class="col-md-12" style="background-color:green">
        col-12
      </div>
      <div class="col-md-6" style="background-color:brown">
        col-6
      </div>
      <div class="col-md-6" style="background-color:red">
        col-6
      </div>
    </div>
  </div>

```

column Demo



```

      col-3
    </div>
    <div class="col-md-3" style="background-color:green">
      col-3
    </div>
    <div class="col-md-3" style="background-color:brown">
      col-3
    </div>
    <div class="col-md-3" style="background-color:red">
      col-3
    </div>
  </div>
</body>

```

Column Example

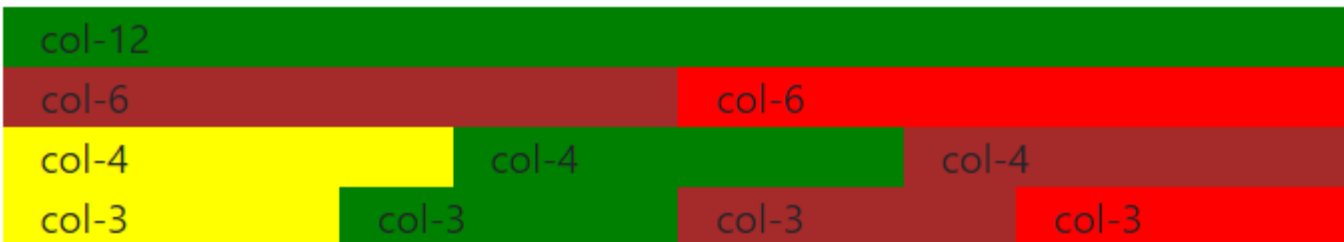
- However, the display will not be maintained on very small display.
- Therefore,
 - Use col-sm property to maintain the original display on small devices.

```

<body>
  <div class="container">
    <h2>column Demo</h2>
    <div class="row">
      <div class="col-sm-12" style="background-color:green">
        col-12
      </div>
      <div class="col-sm-6" style="background-color:brown">
        col-6
      </div>
      <div class="col-sm-6" style="background-color:red">
        col-6
      </div>
      <div class="col-sm-4" style="background-color:yellow">

```

column Demo



```

  </div>
  <div class="col-sm-3" style="background-color:yellow">
    col-3
  </div>
  <div class="col-sm-3" style="background-color:green">
    col-3
  </div>
  <div class="col-sm-3" style="background-color:brown">
    col-3
  </div>
  <div class="col-sm-3" style="background-color:red">
    col-3
  </div>
</div>
</body>

```

Activity

- Create a layout in bootstrap as shown in following figure.

Bootstrap Images

- Bootstrap Images
 - Bootstrap image class provides different image styles.
For example

- Circle



- Rounded Corner



Bootstrap CSS Images Example

- Rounded Corner Images

```
<html lang="en">
<head>
  <link rel="stylesheet" href="bootstrap.min.css">
  <script src="jquery.min.js"></script>
  <script src="bootstrap.min.js"></script>
</head>
<body>

<div class="container">
  <h2>Rounded Corners</h2>
  <p>The .img-rounded class adds rounded corners to an image (not available in IE8):</p>
  
</div>
</body>
</html>
```



Bootstrap Example: Rounded corner img

```
<html lang="en">
<head>
  <link rel="stylesheet" href="bootstrap.min.css">
  <script src="jq
  <script src="bc
</head>
<body>
<div class="conta
  <h2>Rounded Cor
  <p>The .img-rou
  
```

Bootstrap Example

- Alert boxes are used to notify about alert messages. For example

Alerts

Success! This alert box could indicate a successful or positive action.

Info! This alert box could indicate a neutral informative change or action.

Warning! This alert box could indicate a warning that might need attention.

Danger! This alert box could indicate a dangerous or potentially negative action.

Bootstrap Example

```
<html lang="en">
<head>
  <title>Bootstrap Example</title>
  <link rel="stylesheet" href="bootstrap.min.css">
  <script src="jquery.min.js"></script>
  <script src="bootstrap.min.js"></script>
</head>
<body>

<div class="container">
  <h2>Alerts</h2>
  <div class="alert alert-success">
    <strong>Success!</strong> This alert box could indicate a successful or positive action.
  </div>
  <div class="alert alert-info">
    <strong>Info!</strong> This alert box could indicate a neutral informative change or action.
  </div>
  <div class="alert alert-warning">
    <strong>Warning!</strong> This alert box could indicate a warning that might need attention.
  </div>
  <div class="alert alert-danger">
    <strong>Danger!</strong> This alert box could indicate a dangerous or potentially negative action.
  </div>
</div>

</body>
</html>
```

Bootstrap example

- Output

Alerts

Success! This alert box could indicate a successful or positive action.

Info! This alert box could indicate a neutral informative change or action.

Warning! This alert box could indicate a warning that might need attention.

Danger! This alert box could indicate a dangerous or potentially negative action.

Bootstrap example

- List of Bootstrap alert classes

Class	Description
<code>.alert</code>	Creates an alert message box
<code>.alert-danger</code>	Red alert. Indicates a dangerous or potentially negative action
<code>.alert-dismissible</code>	Indicates a closable alert box. Together with the <code>.close</code> class, this class is used to close the alert (adds extra padding)
<code>.alert-info</code>	Light-blue alert. Indicates a neutral informative change or action
<code>.alert-link</code>	Used on links inside alerts to provide matching colored links
<code>.alert-success</code>	Green alert. Indicates a successful or positive action
<code>.alert-warning</code>	Yellow alert. Indicates caution should be taken with this action
<code>.close</code>	Styles the close button for the alert message (floats right with a specified font-size, color, etc.)

Adding NAVBAR

- **Bootstrap Tabs**

```
<!DOCTYPE html>
<html lang="en">

<head>
  <title>Bootstrap Example</title>
  <link rel="stylesheet" href="bootstrap.min.css">
  <script src="jquery.min.js"></script>
  <script src="bootstrap.min.js"></script>
</head>
<body>

<div class="container">
  <h3>Tabs</h3>
  <ul class="nav nav-tabs">
    <li class="active"><a href="#">Home
    <li><a href="#">Menu 1
    <li><a href="#">Menu 2
    <li><a href="#">Menu 3</a></li>
  </ul>
</div>

</body>
</html>
```

Tabs

Home

Menu 1

Menu 2

Menu 3

Adding Pagination to the page

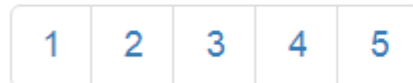
Pagination - Sizing

Add class .pagination-lg for larger blocks or .pagination-sm for smaller blocks.

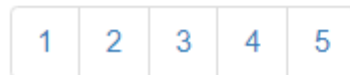
Large:



Default:



Small:



locks.</p>

```
<div class="contai
  <h2>Pagination -
  <p>Add class .pa

<p>Large:</p>
<ul class="pagin
  <li><a href="#"
  <li><a href="#"
  <li><a href="#"
  <li><a href="#"
  <li><a href="#"
</ul>

<p>Default:</p>
<ul class="pagin
  <li><a href="#"
  <li><a href="#"
  <li><a href="#"
  <li><a href="#"
  <li><a href="#"
</ul>

<p>Small:</p>
<ul class="pagin
  <li><a href="#"
  <li><a href="#"
  <li><a href="#"
  <li><a href="#"
  <li><a href="#"></a></li>
</ul>
</div>
```

Adding list to the page

- List Group Items

```
<body>

<div class="container">
  <h2>List Group With Badges</h2>
  <ul class="list-group">
    <li class="list-group-item">New <span class="badge">12</span></li>
    <li class="list-group-item">Deleted <span class="badge">5</span></li>
    <li class="list-group-item">Warnings <span class="badge">3</span></li>
  </ul>
</div>

</body>
```

List Group With Badges

New	12
Deleted	5
Warnings	3

Bootstrap CSS Example

- List group items with col-lg class

```
<div class="container">
  <div class="col-lg-3">
    <h2>List Group With Badges</h2>
    <ul class="list-group">
      <li class="list-group-item">New <span class="badge">12</span></li>
      <li class="list-group-item">Deleted <span class="badge">5</span></li>
      <li class="list-group-item">Warnings <span class="badge">3</span></li>
    </ul>
  </div>
<div class="col-lg-3">
</div>
</body>
```

List Group With Badges

New	12
Deleted	5
Warnings	3

Bootstrap CSS Example

- Checkboxes

- Checkboxes are used if you want the user to select any number of options from a list of preset options.
- The following example contains three checkboxes. The last option is disabled:

Form control: checkbox

The form below contains three checkboxes. The last option is disabled:

- ☒ Option 1
- ☒ Option 2
- ☐ Option 3

Bootstrap CSS Example

```
<!DOCTYPE html>
<html lang="en">

<head>
  <title>Bootstrap Example</title>
  <link rel="stylesheet" href="bootstrap.min.css">
  <script src="jquery.min.js"></script>
  <script src="bootstrap.min.js"></script>
</head>
<body>

<div class="container">
  <h2>Form control: checkbox</h2>
  <p>The form below contains three checkboxes. The last option is disabled:</p>
  <form>
    <div class="checkbox">
      <label><input type="checkbox" value="">Option 1</label>
    </div>
    <div class="checkbox">
      <label><input type="checkbox" value="">Option 2</label>
    </div>
    <div class="checkbox disabled">
      <label><input type="checkbox" value="" disabled>Option 3</label>
    </div>
  </form>
</div>

</body>
</html>
```

Buttons

Use Bootstrap's custom button styles for actions in forms, dialogs, and more with support for multiple sizes, states, and more.

Base class

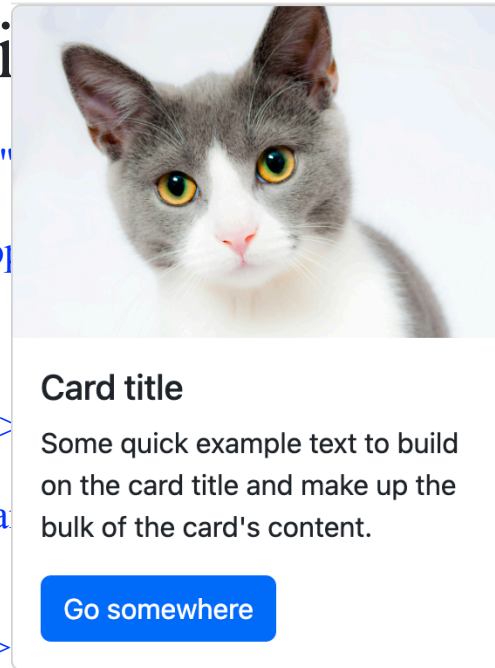
Bootstrap has a base `.btn` class that sets up basic styles such as padding and content alignment. By default, `.btn` controls have a transparent border and background color, and lack any explicit focus and hover styles.

```
<button type="button" class="btn">Base class</button>
```

Cards

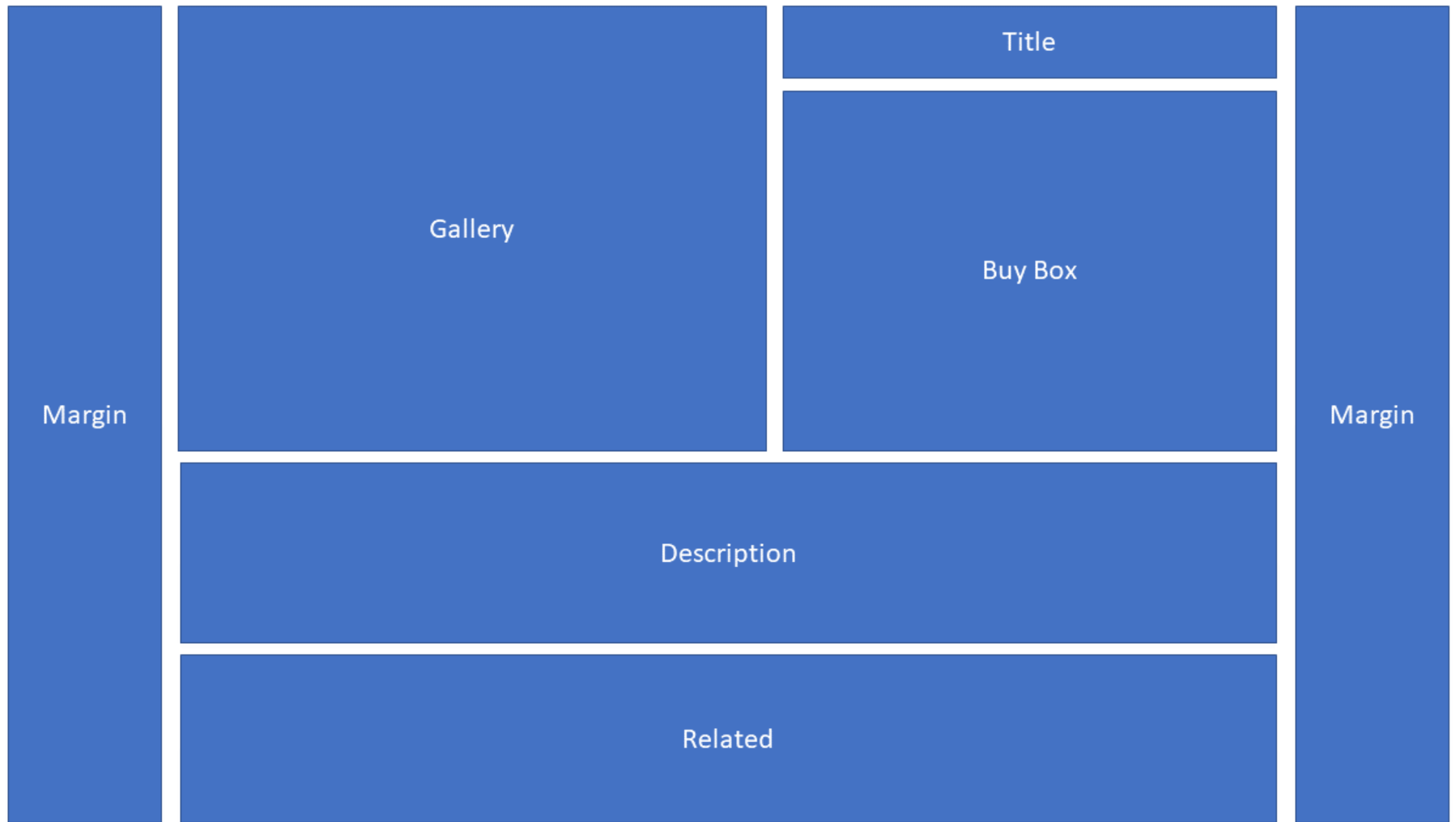
Bootstrap's cards provide a content container with a variety of extendable features and options.

```
<div class="card" style="width: 18rem;">
  
  <div class="card-body">
    <h5 class="card-title">Card title</h5>
    <p class="card-text">Some quick example
content.</p>
    <a href="#" class="btn btn-primary">
  </div>
</div>
```



id make up the bulk of the card's

Bootstrap Grid Example



Bootstrap Grid Example

100 x 60

[Link](#)

[Link](#)

[Link](#)

[Link](#)

600 x 400

300 x 205

300 x 205

300 x 205

300 x 205

300 x 205

300 x 205

300 x 205

300 x 205

600 x 400

100 x 60

100 x 60

60 x 60

30 x 60

100 x 60



Bootstrap Grid Example

